
3.2.2 DNA is an information-carrying molecule. Its sequence of bases determines the structure of proteins, including enzymes.

Structure of DNA	The double-helix structure of DNA, enabling it to act as a stable information-carrying molecule, in terms of • the components of DNA nucleotides: deoxyribose, phosphate and the bases adenine, cytosine, guanine and thymine • two sugar-phosphate backbones held together by hydrogen bonds between base pairs • specific base pairing.
Genes and polypeptides	A gene occupies a fixed position, called a locus, on a particular strand of DNA. Genes are sections of DNA that contain coded information as a specific sequence of bases. Genes code for polypeptides that determine the nature and development of organisms. A sequence of three bases, called a triplet, codes for a specific amino acid. The base sequence of a gene determines the amino acid sequence in a polypeptide. In eukaryotes, much of the nuclear DNA does not code for polypeptides. There are, for example, introns within genes and multiple repeats between genes. Differences in base sequences of alleles of a single gene may result in non-functional proteins, including non-functional enzymes.
DNA and chromosomes	In eukaryotes, DNA is linear and associated with proteins. In prokaryotes, DNA molecules are smaller, circular and are not associated with proteins.
Meiosis	The importance of meiosis in producing cells which are genetically different. Within this unit, meiosis should be studied only in sufficient detail to show • the formation of haploid cells • independent segregation of homologous chromosomes. Gametes are genetically different as a result of different combinations of maternal and paternal chromosomes • genetic recombination by crossing over.
